

AIDAN SPARLING

Software Engineer – Frontend & Systems

Willing to relocate to El Segundo, CA
276-210-0535 | aidansparling@gmail.com

SUMMARY

Software engineer with hands-on experience building data-driven web applications, real-time dashboards, and embedded control systems. Strong background bridging software, and hardware. Passionate about infrastructure, energy, and intuitive visualization of complex data. Strong interest in nuclear energy.

TECHNICAL SKILLS

Frontend: JavaScript, Django, HTML/CSS, Data Visualization

Backend & Data: Python, C#, C, Ruby, Java, SQL, REST APIs

Systems & Infra: Docker, Git, Linux

Embedded & Controls: CAN Bus, Real-Time Systems, Sensor Telemetry, Closed-Loop Control

EXPERIENCE

IT Intern — Granges, Huntingdon, TN
Summer 2025

- Developed an internal **service logging application** using **C# and SQL**, improving visibility into system issues and maintenance workflows
 - Collaborated with cross-functional teams to maintain and update **software and hardware systems** supporting production operations
 - Diagnosed and resolved system failures with a focus on reliability, uptime, and clear communication
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ENGINEERING PROJECTS

Custom Engine Management System

- Designed and implemented a **fully custom engine management system** on a legacy vehicle platform
- Built a **bespoke wiring harness** and integrated ECU-controlled **fully sequential fuel and spark**
- Implemented **flex-fuel logic**, dynamically adjusting parameters based on real-time sensor data
- Designed and tuned **closed-loop cruise control** using feedback control principles
- Validated system behavior using live telemetry, logging, and iterative real-world testing

Macro & Health Tracking Web Application

- Built a **personalized health tracking platform** generating user-specific recommendations from individual data
 - Integrated the **USDA FoodData Central (FDC) API** to ingest detailed nutritional data for meal logging
 - Designed **interactive dashboards and visualizations** to display macro and nutrient trends over time
 - Implemented intuitive workflows connecting frontend UI with backend services and persistent user data
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EDUCATION

B.S. Computer Science — James Madison University
December 2025

Relevant Coursework: Algorithms & Data Structures, Software Engineering, Systems Programming